Program : **Diploma in Architecture**	
Course Code : **4182**	Course Title: **Surveying for Architecture**
Semester : **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4** (**L:3, T:1, P:0**)	Periods per semester: **60**

Course Objectives:

- To introduce types of surveying and equipment to be used for different surveys.
- To provide knowledge about plane table surveying and The odolite surveying.
- To introduceworking principles of Electronic Distance Measurement equipment and Total station and their uses.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic knowledge about mathematics		Engineering Mathematics	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Select the type of survey suitable for given situation	12	Applying
CO2	Use levelling instruments to determine reduced level for preparation of contour maps	17	Applying
CO3	Use theodolite to determine area of the field	15	Applying
CO4	Make use of modern survey instruments and understand the application of remote sensing and GIS in the field of architecture	14	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3		2	3	3		
CO3	3		2	3	3		
CO4	3		2	3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Select the type of survey suitable for given situation**		
M1.01	Explain principle and types of surveying used in various fields	4	Understanding
M1.02	Compute area of open field using chain, tape and cross staff	4	Applying
M1.03	Illustrate traversing in field using chain and compass	4	Applying
Contents: Purpose of surveying - types of survey - chain survey - calculate areas - functions of chain, tape and cross - staff - operations involved in chain survey to make site plan - survey plan from given field notes - principles of plane table survey - functions of accessories of plane table - operations to set up and orient the plane table - principle of Compass survey - parts and their functions of a prismatic compass-operations involved in the field to take various bearings			
CO2	**Use levelling instruments to determine reduced level for preparation of contour maps**		
M2.01	Explain the principle of levelling and different types of leveling	5	Understanding
M2.02	Solve leveling problems and determine the reduced levels and plot contours.	12	Applying
	Series Test I	1	
Contents: Principle of leveling - parts and functions of dumpy level and telescopic leveling staff - steps involved in performing the temporary adjustments of leveling Instrument - record observations in field book - compute the reduced levels of stations from field book - Classifications of leveling - principle of contouring - contour plans from given field notes			

CO3	Use theodolite to determine area of the field		
M3.01	Describe the parts and function of theodolite	4	Understanding
M3.02	Explain the principle of theodolite traversing	5	Understanding
M3.03	Identify the methods used for traverse survey using theodolite to calculate area	6	Applying
Contents: Principles of theodolite traversing - their parts and function -temporary adjustments and terms - fundamental lines and their relationship - horizontal angle measurement by repetition and reiteration method - steps involved in setting out angles using a theodolite - method of conducting traverse survey in the field using the theodolite - compute the co-ordinates			
CO4	**Make use of modern survey instruments and understand the application of remote sensing and GIS in the field of architecture**		
M4.01	Explain modern survey instruments	4	Understanding
M4.02	Define the Application of GIS in architecture	4	Understanding
M4.03	Determine height, horizontal distances and prepare drawings using total station at field	6	Applying
	Series Test II	1	
Contents: Modern surveying equipment - electronic theodolite - distomat - laser level - GPS - remote sensing in architecture - application of GIS in architecture - principles of Total Station - features and technical terms - method of operation - contour map of an area - Determine height of building - horizontal distance between points - data downloading.			

Text /Reference:

T/R	BookTitle/Author
T1	Punmia, B.C Jain, Ashok Kumar; Jain, Arun Kumar, Surveying I, Laxmi Publications, New Delhi.
R2	Kanetkar, T. P.; Kulkarni, S. V., Surveying and Levelling volume I, Pune Vidyarthi Gruh Prakashan.
R3	Duggal, S. K., Survey I, McGraw Hill Education, New Delhi.
R4	Saikia, M D.; Das. B.M.; Das. M.M., Surveying, PHI Learning, New Delhi.
R5	Subramanian, R., Fundamentals of Surveying and Levelling, Oxford University Press. New Delhi.
R6	Rao, P. Venugopala Akella, Vijayalakshmi, Textbook of Surveying, PHI Learning New Delhi.
R7	Arora K R, Surveying Vol. I, Standard Book House
R8	Bhavikatti, S. S., Surveying and Levelling, Volume 1, I. K. International, New Delhi.

Online resources:

Sl.No	Website Link
1	https://lecturenotes.in/subject/156/surveying-1-s-1
2	https://civiltoday.com/surveying/87-surveying-lecture-notes-pdf
3	http://www.a-zshiksha.com/forum/viewtopic.php?f=149&t=61476
4	https://www.smartzworld.com/notes/surveying-pdf-notes-lecture-notes-free-
5	https://edurev.in/studytube/Surveying--Part-1--Introduction-Notes--Surveying--